# RetailDB Service

1.0.0 OAS3

/openapi.yaml

API for the RetailDB coursework service

Servers

/

Authorize

## default

**POST** `/auth/login` Obtain JWT for admin

**Parameters** Try it out

No parameters

**Request body** required

application/json

**Examples:**

valid

**Example Value** Schema

```
{
  "username": "admin",
  "password": "password"
}
```

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Token returned | *No links* |

| Code | Description | Links |
|---|---|---|
| | Media type: application/json (Controls Accept header.) Examples: sample<br>Example Value / Schema<br>`{ "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..." }` | |
| 400 | Bad request (invalid JSON) | *No links* |
| 401 | Unauthorized (invalid credentials) | *No links* |
| 500 | Internal server error | *No links* |

**GET** `/healthz` Health check

**Parameters** Try it out

No parameters

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | OK<br>Media type: application/json (Controls Accept header.) Examples: ok<br>Example Value / Schema<br>`{ "status": "ok"` | *No links* |

| Code | Description | Links |
|---|---|---|
| | } | |

**GET** `/products` List products

**Parameters** Try it out

| Name | Description |
|---|---|
| limit<br>`integer`<br>*`(query)`* | *Default value* : 20<br>20 |
| offset<br>`integer`<br>*`(query)`* | *Default value* : 0<br>0 |

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | A list of products<br><br>Media type: application/json<br>Controls `Accept` header.<br>Examples: sample<br><br>**Example Value** Schema | *No links* |

```
[
  {
    "product_id": "AH2430",
    "product_name": "None",
    "brand": "None",
    "description": "None",
    "last_visited": "2018-05-19 15:13:00"
  },
  {
```

| Code | Description | Links |
|---|---|---|


```
    "product_id": "G27341",
    "product_name": "Women's adidas Originals Sleek Shoes",
    "brand": "Adidas",
    "listing_price": 76,
    "sale_price": 38,
    "discount": 0.5,
    "revenue": 1641,
    "description": "A modern take on adidas sport heritag
e...",
    "reviews": 24,
    "last_visited": "2018-11-29 16:16:00"
  }
]
```


| | | |
|---|---|---|
| 400 | Bad request (invalid query parameters) | *No links* |
| 500 | Internal server error | *No links* |

**GET** `/products/{id}` Get product by ID

**Parameters** Try it out

| Name | Description |
|---|---|
| **id** * required<br>`string`<br>***(path)*** | id |

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Product<br><br>Media type: **application/json** (Controls `Accept` header.)<br>Examples: **sample** | *No links* |

| Code | Description | Links |
|---|---|---|

**Example Value** Schema


```
{
  "product_id": "G27341",
  "product_name": "Women's adidas Originals Sleek Shoes",
  "brand": "Adidas",
  "listing_price": 76,
  "sale_price": 38,
  "discount": 0.5,
  "revenue": 1641,
  "description": "A modern take on adidas sport heritage...",
  "rating": 4.2,
  "reviews": 24,
  "last_visited": "2018-11-29 16:16:00"
}
```


| | | |
|---|---|---|
| 400 | Bad request (invalid ID format) | *No links* |
| 404 | Not found | *No links* |
| 500 | Internal server error | *No links* |

**POST** `/query` AI-powered natural language database query

Send a natural-language question and the service uses Gemini to generate a read-only SQL SELECT query, executes it against the database, and returns both the generated SQL and the result rows. Requires the GEMINI_API_KEY environment variable to be set on the server.

**Parameters** Try it out

No parameters

**Request body** required application/json

**Examples:**

sample

**Example Value** Schema


```
{
```

```
  "question": "What are the top 5 most expensive products?"
}
```

## Responses

| Code | Description | Links |
|---|---|---|
| 200 | Query executed successfully<br><br>Media type: application/json<br>Controls Accept header.<br>Examples: sample<br><br>**Example Value** Schema | *No links* |
| 400 | Bad request (missing or empty question) | *No links* |
| 500 | Query generation or execution failed<br><br>Media type: application/json<br><br>**Example Value** Schema | *No links* |

```
{
  "sql": "SELECT i.product_id, i.product_name, f.listing_pric
e FROM info i LEFT JOIN finance f ON i.product_id = f.produc
t_id ORDER BY f.listing_price DESC LIMIT 5",
  "results": [
    {
      "product_id": "G27341",
      "product_name": "Women's adidas Originals Sleek Shoe
s",
      "listing_price": 76
    }
  ]
}
```

```
{
  "error": "string",
  "sql": "string"
}
```

| Code | Description | Links |
|---|---|---|
| 503 | LLM query not configured (GEMINI_API_KEY missing) | *No links* |

**POST** `/admin/products` Create a new product

**Parameters** Try it out

No parameters

**Request body** required application/json

**Examples:**

sample

**Example Value** Schema

```
{
  "product_id": "NEW1",
  "product_name": "My Product",
  "brand": "Acme",
  "listing_price": 12.34,
  "sale_price": 9.99,
  "discount": 0.19,
  "revenue": 99,
  "description": "Short description",
  "rating": 4.1,
  "reviews": 12,
  "last_visited": "2022-01-01 00:00:00"
}
```

**Responses**

| Code | Description | Links |
|---|---|---|
| 201 | Created<br>Media type: application/json<br>Controls Accept header.<br>Examples: ok | *No links* |

| Code | Description | Links |
|---|---|---|
| | **Example Value** Schema<br>`{ "status": "created", "product_id": "NEW1" }` | |
| 400 | Bad request (invalid payload) | *No links* |
| 401 | Unauthorized (missing or invalid token) | *No links* |
| 409 | Conflict (product already exists) | *No links* |
| 500 | Internal server error | *No links* |

**PUT** `/admin/products/{id}` Update modified fields for a product

**Parameters** Try it out

| Name | Description |
|---|---|
| **id** * required<br>`string`<br>*(path)* | id |

**Request body** required application/json

**Examples:**

update-example

**Example Value** Schema

```
{
  "modified_product_name": "New Name",
  "modified_description": "Updated description",
  "modified_brand": "MyBrand",
  "modified_listing_price": 99.99
}
```

### Responses

| Code | Description | Links |
|---|---|---|
| 200 | Updated<br>Media type: application/json<br>Controls Accept header.<br>Examples: ok<br>Example Value Schema<br>`{ "status": "updated" }` | *No links* |
| 400 | Bad request (invalid update payload) | *No links* |
| 401 | Unauthorized (missing or invalid token) | *No links* |
| 403 | Forbidden (insufficient privileges) | *No links* |
| 500 | Internal server error | *No links* |

**DELETE** `/admin/products/{id}` Delete product by ID

### Parameters

Try it out

| Name | Description |
|---|---|
| **id** * required<br>`string`<br>*(path)* | id |

## Responses

| Code | Description | Links |
|---|---|---|
| 200 | Deleted<br><br>Media type: application/json<br>Controls Accept header.<br>Examples: ok<br><br>Example Value Schema<br><br>{<br>"status": "deleted",<br>"product_id": "DEL1"<br>} | *No links* |
| 401 | Unauthorized | *No links* |
| 404 | Not found | *No links* |
| 500 | Internal server error | *No links* |

## Schemas

### LoginRequest

### LoginResponse

### Product

### ProductCreate

**UpdateModified**

**QueryRequest**

**QueryResponse**